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Age, Paleomagnetism, and Tectonic Significance of the Elberton Granite, Northeast Georgia Piedmont

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The Rb/Sr whole rock age of 350 ± 11 m.y., the remanent magnetism (RM), and petrology of samples from 21 sites within the granitic Elberton pluton (NE Georgia) have been determined. Two partially conflicting estimates of tectonic rotation are developed and evaluated. The first, based on the orientation of the magnetic foliation within the body and its reorientation, assuming an originally horizontal upper granite contact and coincident magnetic fabric, requires a post emplacement rotation of 30°–35° down to the SE about a N-NE axis. The second, based on reorienting the body so that the pole determined for the granite before tilt correction ($\phi = 120.3$; $\theta = 54.7$; $d_p = 2.2$; $dm = 4.5$) becomes coincident with an independently reported mean 350 m.y. B.P. paleopole ($\phi = 120$; $\theta = 47$), indicates a postemplacement rotation for the body of ~15° down to the NW along a E-NE axis. While these tilt estimates are different, they do have two very important points in common, namely, that the strike of rotation is similar and the magnitude of rotation is small. We conclude from these data that no major structural reorientation or relocation of the Elberton granite is necessary to produce a pole position consistent with the 350 m.y. Rb/Sr whole rock age. We suggest that the tilting of the granite (if any) that may have occurred resulted from uplift of the Inner Piedmont, with the Elberton pluton isolated within a rigid crustal block bounded by NE-SW trending faults moving in response to that uplift. Our interpretation of these data is not consistent with a post 350 m.y. major continental collision event, with concurrent large-scale thrust fault motion, involving the Southern Appalachians of North America. This conclusion is reinforced by tectonic analyses based on the regional gravity gradient and on oxygen isotopic data reported for late orogenic granites in the Southern Piedmont.

INTRODUCTION

The Elberton granite is a large, fine-grained pluton intruded into orthogneisses and paragneisses of the Inner Piedmont of eastern Georgia (Figure 1). It is the only major pluton cut by the recent seismic line across the Southern Appalachians run by the Consortium for Deep Continental Reflection Profiling (Cocorp) [*Cook et al.*, 1979]. This seismic data has been interpreted as demonstrating that Inner Piedmont rocks terminate at a depth of about 10 km and are underlain by flat-lying reflectors which are interpreted as equivalents of the Valley and Ridge sedimentary rocks. Thus the age and tectonic history of the Elberton granite are important in interpreting the history of the Southern Appalachians. The current study represents the second phase of a study of the paleomagnetic properties of this body [*Ellwood and Whitney*, 1980] to determine a magmatic and tectonic history for the unit.

GEOCHRONOLOGY

Previous work. The first Elberton radiometric data reported are $^{238}U/^{206}Pb$, $^{235}U/^{207}Pb$, and $^{207}Pb/^{206}Pb$ ages from zircons [*Grunenfelder and Silver*, 1958]; the data are also reported by *Fairbairn et al.* [1960] and *Butler* [1972]. These age determinations correspond to Ordovician time (ages of about 450 to 490 m.y.). However, it appears that an Ordovician age for the Elberton is inconsistent with the structural relationships and uplift history of the Piedmont [*Dallmeyer*, 1978]. We now believe that these ages are greater than the crystallization age of the Elberton, and we suspect that many of the Elberton zircons are inherited and were not 'reset' (recrystallized or greatly overgrown) by the low magmatic temperatures of the Elberton. Similar conclusions about U/Pb ages from other Piedmont plutons have been reached by *Higgins et al.* [1977].

K/Ar data using mica from the Elberton have yielded ages (235 and 245 m.y.) which correspond to Permian time [*Fairbairn et al.*, 1960; *Smith et al.*, 1969]. These ages are now interpreted to be uplift ages [*Dallmeyer*, 1978], recording the time when the Elberton cooled to a temperature of about 300° to 350°C.

Methods. Fifteen samples of about 10 kg each were collected from the Elberton pluton. Each sample was crushed and split to a 10-g portion, which was subsequently powdered. The powder was split into a 0.2 to 0.3 g portion for Rb and Sr isotopic analysis. Each analysis was done using ^{84}Sr and ^{87}Rb spikes, ultrapure HF, $HClO_4$ and HCl, and Teflon beakers. The solutions were passed through pyrex cation exchange resin columns to obtain Rb and Sr fractions.

The Sr isotopic analyses were made using a Nier-type, 6-inch radius mass spectrometer (SIMA) with a programable automatic data acquistion system at the Department of Ter-

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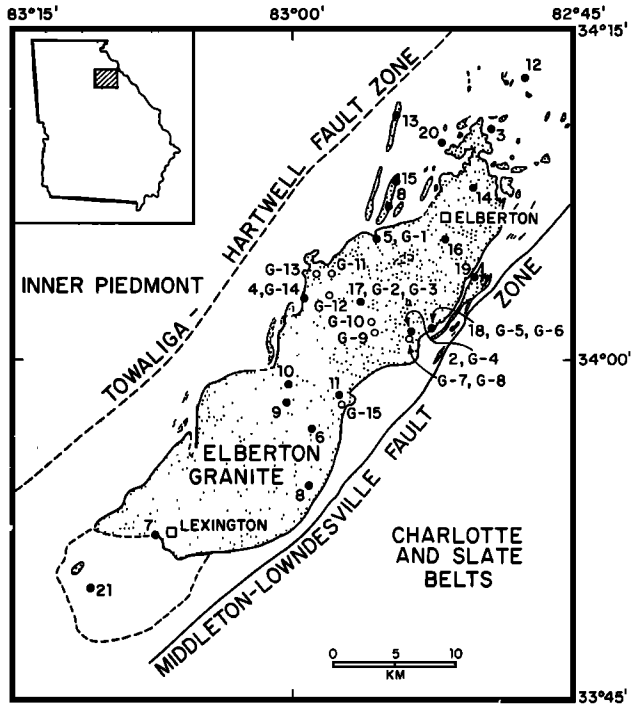


Fig. 1. Location of the Elberton Granite (stippled), Georgia, in relationship to the Inner Piedmont, Charlotte–Slate Belts, and shear zones (diagonals). Dashed line outlines the older Hutchin metagranite. Sample sites are numbered. Closed circles are paleomagnetic sites; open circles are Rb/Sr and $\delta^{18}O$ sites.

restrial Magnetism of the Carnegie Institute. The Rb isotopic analyses were made using a Nier-type 12-inch radius mass spectrometer at Florida State University.

All the Sr isotopic compositions were calculated from analyses of sample plus spike mixtures. The $^{85}Rb/^{87}Rb$ ratio was taken to be 2.593 and the decay constant used for ^{87}Rb is 1.42×10^{-11} yr^{-1} [*Steiger and Jager,* 1977]. The Rb/Sr age and initial $^{87}Sr/^{86}Sr$ ratio on the whole rock isochron diagrams were calculated using the *York* [1966] regression treatment. The one-standard-deviation experimental error in $^{87}Rb/^{86}Sr$ was derived from an examination of duplicate analyses done over the past 6 years and was calculated to be 2%. The one-standard-deviation experimental error in $^{87}Sr/^{86}Sr$ was derived from an examination of multiple analyses of the Eimer and Amend standard $SrCO_3$ made during the time that these Sr analyses were performed and from an examination of the errors obtained from the mass spectrometer data. The one-standard-deviation experimental error of the 10 Eimer and Amend analyses was 0.005%; the experimental error in the normalized $^{87}Sr/^{86}Sr$ ratio for each sample analysis was acceptable when less than 0.02%. The one-standard-deviation experimental error for $^{87}Sr/^{86}Sr$ used to calculate the Rb/Sr isochron age was 0.05%. The data points on the whole rock isochron diagrams are shown with error boxes whose dimensions correspond to two standard deviations. The errors assigned to the isochron age and the calculated initial $^{87}Sr/^{86}Sr$ ratio on the isochron diagrams are given at the 68% confidence level (1 σ). The analytical data and geological setting are presented in Table 1.

Discussion. Nine of the samples (4, 5, 6, 7, 8, 10, 12, 13, and 14) came from rock quarry sites that showed no apparent problems in the form of nearby xenoliths, hydrothermal alteration, or cataclastic zones. These nine samples yield an isochron age of 350 ± 11 m.y. and an initial $^{87}Sr/^{86}Sr$ ratio of 0.7037 ± 0.0005 (solid line in Figure 2). Three of the collected samples (1, 2, and 9) came from quarry sites in which there were abundant xenoliths, though almost all the xenoliths appeared to have sharp angular boundaries and exhibited no obvious evidence of assimilation. One sample (sample 3) came from a pegmatite dike within the Elberton. Another sample (sample 11) came from a site near an apparent hydrothermally formed vug, 1 cm in diameter, possibly containing copper mineralization. Still another sample (sample 15) came from a relatively distant locality in the Vesta 7½-foot quadrangle, south of the Elberton West 7½-foot quadrangle in which all the other samples were collected.

The samples with possible problems (1, 2, 3, 9, 11, and 15) were not used to determine the Rb/Sr whole rock age of the Elberton. However, it proved interesting to note that analyses of these samples fall on an isochron with a calculated age of 376 ± 45 m.y. and an initial $^{87}Sr/^{86}Sr$ of 0.7054 ± 0.0024 (dashed line in Figure 2). Although this isochron age and initial $^{87}Sr/^{86}Sr$ ratio are essentially the same within the analytical errors, as the age and initial $^{87}Sr/^{86}Sr$ ratio derived from the samples which have no apparent problems, a composite of these data shows that at the 95% confidence level the sample analyses are better fit to two different isochrons than to a single isochron. We now believe that the 350 ± 11 m.y. isochron derived from the samples without apparent problems is more meaningful in terms of determining the age of the Elberton. The fact that the isochron derived from anomalous samples, represented by the dashed line in Figure 2, lies slightly above the isochron derived from the main body of the Elberton may be due either to ^{87}Sr enrichment caused by some subtle form of contamination due to the incorporation of radiogenic ^{87}Sr from the country rock (with no obvious evidence of assimilation of the xenoliths by the magma except at one locality, i.e., location 9) or to heterogeneities in the initial $^{87}Sr/^{86}Sr$ of the magma. In any case, plutonic rocks from the Appalachians (and elsewhere) that yield what have been called parallel isochrons have been reported by others [*Spooner and Fairbairn,* 1970; *Fullagar and Butler,* 1976; *Butler and Fullagar,* 1978; *Mose and Wenner,* 1980].

Samples 9 and 14 were arbitrarily selected for a Rb/Sr mineral isochron study (Table 1). Biotite–whole rock isochrons, for these samples, yield ages of 266 ± 6 m.y. for sample 9 and 276 ± 6 m.y. for sample 14, errors calculated assuming an analytical error of 2% and 0.05% in $^{87}Rb/^{86}Sr$ and $^{87}Sr/^{86}Sr$, respectively. Earlier Rb/Sr studies using mica extracted from Elberton granite samples yielded ages of 248 ± 13 m.y. from biotite and 240 ± 13 m.y. from muscovite (ages recalculated from *Pinson et al.* [1958] and *Fairbairn et al.* [1960], using the ^{87}Rb decay constant = 1.42×10^{-11} yr^{-1}). Although these Rb/Sr mica ages and the earlier mentioned K/Ar mica ages of about 235 and 245 m.y. are about 100 m.y. younger than the Rb/Sr whole rock age (and about 200 m.y. younger than the U/Pb ages), these mica ages are not thought to represent a time of metamorphism, since the granite contains no metamorphic fabric. Instead, these mica ages are assumed to be related to the time at which the pluton cooled during uplift to a temperature below which radiogenic argon and radiogenic strontium were retained in the dated micas.

The long interval between the time of granite crystallization and the time of cooling to the retention temperature of mica seen in the Elberton has an implication with regard to the interpretation of Rb/Sr data. In earlier studies the absence of

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TABLE 1. Rb/Sr and $^{18}O/^{16}O$ Analyses of Elberton Samples

Sample	Measured $^{87}Rb/^{86}Sr$ (atomic ratio)	Measured $^{87}Sr/^{86}Sr$ (atomic ratio)	Error, %	^{86}Sr, ppm	^{87}Rb, ppm	$\delta^{18}O$,* Smow	Geologic Setting
G 1	4.488	0.7920	0.015	15.866	72.036	7.9 ± 0.10	few xenoliths in area
G 1a	4.500	0.7282	0.014	15.813	71.983		
Average	4.494	0.7286					
G 2	3.730	0.7263	0.019	17.788	67.126	8.5	many xenoliths in area
G 3	3.815	0.7256	0.015	12.974	50.069	8.5	pegmatite dike sample
G 4†	3.736	0.7232	0.005	19.010	71.846	7.4 ± 0.16‡	no xenoliths visible
G 5†	4.904	0.7277	0.016	15.446	76.620	7.7	no xenoliths visible
G 5a	4.874	0.7274	0.015	15.780	77.798		
Average	4.889	0.7276					
G 6†	4.869	0.7271	0.012	16.004	78.828	7.6	no xenoliths visible
G 6a	4.898	0.7275	0.013	15.806	78.313		
Average	4.883	0.7273					
G 7†	4.142	0.7314	0.020	17.088	71.597	7.6	no xenoliths visible
G 8†	4.142	0.7530	0.014	15.702	66.210	7.6	aplite with soft margins, coeval with magma
G 9	2.984	0.7215	0.012	20.723	62.651	8.0 ± 0.02	many xenoliths in area
G 9 Biotite	137.700	1.2303	0.014	2.001	278.660		
G 10†	1.438	0.7107	0.009	32.006	46.549	6.8 ± 0.03	no xenoliths visible
G 11	3.498	0.7233	0.012	18.799	66.525	8.4 ± 0.04	vug near sample site; gneissic (plastic deformation) zone about 30 m away
G 12†	1.475	0.7116	0.180	36.107	53.885	7.7	some gneissic rock in area but located far from sample site
G 13†	2.531	0.7175	0.014	26.599	63.254	7.7 ± 0.05	no xenoliths visible
G 14†	1.516	0.7110	0.020	31.191	47.833	7.0	no xenoliths visible
G 14 Biotite	42.599	0.8839	0.020	4.886	210.530		
G 15	4.305	0.7289	0.013	16.150	70.332	8.8	collected some distance to south in the Vesta quadrangle

*No ± indicates one analysis; ± indicates average deviation of two samples. All samples are reported relative to the Smow standard in which NBS standard 28 is regarded to be +9.61.

†Samples used to define the Elberton age isochron. Sample locations are given in Figure 1.

‡Represents average and standard deviation of 10 separate samples collected over an area of ~8000 m^2.

metamorphic fabric in a pluton has been taken to mean that a Rb/Sr mica–whole rock isochron will reveal the time of crystallization. This is based on the assumption that plutons cool quickly, and this assumption has been used to interpret Rb/Sr mica–whole rock isochron ages as representing the time of crystallization (Stone Mountain pluton from *Whitney et al.* [1976]; Clouds Creek, Coronaca, and Churchland plutons from *Fullagar and Butler* [1979]. Rb/Sr mineral–whole rock isochrons clearly yield cooling ages younger than the time of crystallization, and as shown by the Elberton study, the crystallization age may be considerably greater than the cooling age. As pointed out by *Fullagar and Butler* [1979], the initial $^{87}Sr/^{86}Sr$ ratio of the mineral isochron may be helpful in evaluating the difference between the cooling age and the crystallization age.

Supporting $^{18}O/^{16}O$ data. All of the samples collected for whole rock Rb/Sr analyses were also analyzed for their $^{18}O/^{16}O$ ratios. These data are reported in Table 1. Many of the sample localities which were collected for Rb/Sr age dating had previously been analyzed for their oxygen isotopic compositions in order to delineate possible variations in the initial $^{87}Sr/^{86}Sr$ of the magma. Such a possibility was anticipated, since it has been shown in many studies that a correlation often exists between the $^{18}O/^{16}O$ and the initial $^{87}Sr/^{86}Sr$ of many plutonic and volcanic complexes (see, for example, *O'Neil and Chappell* [1977] and *Taylor and Silver* [1978]). Further, it was expected that the Elberton Pluton might exhibit variations in its initial strontium isotopic composition, despite its extreme chemical, mineralogical, and textural uniformity, since previous studies revealed a rather broad $^{18}O/^{16}O$ variation [*Wenner et al.*, 1977]. In fact, such correlations were observed among the different late orogenic plutons in the Southern Piedmont [*Wenner et al.*, 1977].

The oxygen isotopic compositions of the samples, presented in Table 1, reveal a broad spectrum of $\delta^{18}O$ values that vary from 6.8 to 8.8. This variation encompasses most of the total range of oxygen isotopic values (6.1 to 9.3; *Wenner* [1980]) encountered among the 42 samples analyzed to date from this pluton.

The samples that define the two parallel Rb/Sr isochrons shown in Figure 2 have two distinct $\delta^{18}O$ ranges; samples that

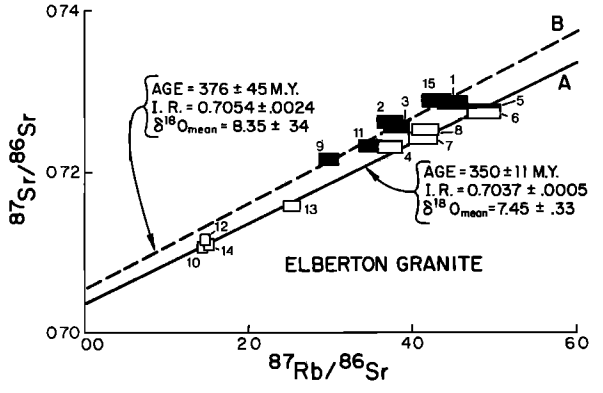


Fig. 2. Composite Rb/Sr isochron diagram showing the whole rock sample analyses from problem free locations (open error boxes) and from possible problem locations (filled error boxes).

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define the lower isochron range from 6.8 to 7.7, with a mean of 7.45 ± 0.33, whereas those that form the upper isochron are ^{18}O enriched, varying from 7.9 to 8.8, with a mean of 8.35 ± 0.34.

Although the $^{18}O/^{16}O$ range (and correspondingly, the initial $^{87}Sr/^{86}Sr$ variations) can not be unambiguously interpreted, it seems clear that such variations are not due to (1) secondary processes (there is no petrographic or isotopic evidence of disequilibrium between coexisting minerals; *Wenner et al.* [1977]), (2) variations in model or chemical compositions which are exceedingly small [*Hess and Stormer,* 1980], and (3) conspicuous assimilative effects, despite the general occurrence of xenoliths located in sample localities that are commonly ^{18}O enriched (see Table 1). In general, there appears to be no obvious chemical or geologic evidence of assimilation of any xenoliths by the Elberton magma except at one sample site (see Table 1). Virtually all xenoliths are composed of previously metamorphosed high-grade gneisses, schists, and minor calcsilicate rocks that display sharp contacts with the granite. There appear to be no oxygen isotopic gradients within the granite adjacent to such xenoliths or to overlying country rock [*Wenner,* 1980].

It can tentatively be concluded that the $^{18}O/^{16}O$ and initial $^{87}Sr/^{86}Sr$ variations observed within the Elberton batholith must be due either (1) to some subtle, but relatively broad scale (of the order of ~8000 m²) contamination effect, which would only be manifested by variations in $^{18}O/^{16}O$ and $^{87}Sr/^{86}Sr$ and some dispersed trace elements [*Hess and Stormer,* 1980] but not by major element changes or (2) to initial isotopic and trace element variations within the protolith that were in turn reflected by the magma during partial anatexis.

PALEOMAGNETIC AND PETROLOGIC STUDIES

Methods

Field. A total of 266 cores were taken with a gasoline-powered portable drill from 20 quarries and one blasted road cut in the Elberton Pluton (Figure 1) and oriented using a Brunton Compass. Core sites were selected which avoided (1) rocks within 50 cm of surfaces cut by a flame cutting tool during quarrying, (2) weathered surfaces, and (3) any unusual features such as xenoliths, pegmatites, or sheared surfaces. Cores were separated laterally by an average distance of greater than 10 m.

Laboratory. The cores, of ~2.5-cm diameter, were sliced into lengths of ~2.2 cm to conform to an optimum length to diameter ratio range of 0.82–0.90 [*Banerjee and Stacey,* 1967; *Noltimier,* 1971]. The natural remanent magnetization (NRM) was measured for all samples by using a Schonstedt spinner magnetometer.

To isolate and identify the characteristic or primary magnetization in pilot samples, a stepwise af demagnetization was performed on the first sample taken from all but one site. Samples were demagnetized, after NRM measurement, at 10 mT steps to 110 mT for the first six sites and at 20 mT steps for remaining sites. The data were then analyzed by using standard techniques, including *J/Jo* plots, declination-inclincation equal area plots, and methods presented by *Symons and Stupavsky* [1974] and *Stupavsky and Symons* [1978]. Due to the wide apparent range of coercivities in the granite (Table 2), the remanent magnetization (RM) of remaining samples was measured after demagnetization to 40 mT and 80 mT. Sixteen samples from several sites were selected for microscopic analysis, and polished sections were prepared and examined under reflected light.

TABLE 2. Elberton Pilot Sample af Demagnetization Data

Site	PSI	Optimum Field	Model
1	225.6	50	4
2	60.1	60	5
3	4.2	80	4
4	8.9	70	4
5	157.0	40	4
6	129.5	20	5
7	13.5	60	5
8	15.4	80	5
9	2.8	60	4
10	5.8	100	4
11	5.5	60	4
12	189.3	20	4
13	42.5	80	4
14	5.6	60	4
15	102.1	60	3
16	2.4	100	2
17	1.5	100	4
18	8.8	100	4
19	4.5	100	4
20	44.8	100	4

PSI is paleomagnetic stability index of *Symons and Stupavsky* [1974]; optimum field in milliteslas predicted by the PSI; model number predicted by the separation of magnetic components method of *Stupavsky and Symons*[1978], where 2 is an exponential primary component of magnetization plus a constant vector, 3 is an exponential primary and secondary component of magnetization, 4 is exponential primary and secondary component plus a constant vector, and 5 is an exponential primary and two secondary components of magnetization.

A selected suite of samples was thermally demagnetized at 100°C intervals by using a Schonstedt thermal demagnetizer, and the RM was measured after each run on a superconducting technology cryogenic magnetometer. Thermal as well as af demagnetization was performed to isolate spurious magnetic moments [*Parry,* 1967]. A single sample was thermally demagnetized while it remained within the cryogenic magnetometer, and the change in magnetization was monitored. Several Curie temperatures were determined with a thermomagnetic balance for chips placed in a 300 mT magnetic field. Measurements were made in vacuo.

Results

RM results for pilot samples are given in Table 2. In Table 3 are presented RM data after demagnetization at 40 mT for initially measured samples from all 21 sites. Data from each site were evaluated for internal consistency, and anomalous directions (asterisks in Table 3) were not considered in calculating the site mean directions given in Table 4. The eliminated samples either represent anomalous polarities within an otherwise unipolarity site or an attempt to improve the precision of the data. One site, EB 12, was rejected based on poor overall precision for the site ($\alpha_{95} > 30°$).

Curie temperatures for samples analyzed ranged from 564°–580°C, indicating that magnetite is present. No evidence of ilmeno-hematite or ferrianilmenite Curie temperatures was apparent. This may be due to the dominance of the magnetite saturation remanence. Examination of polished sections from several sites reveals that the dominant iron oxide is an ilmenite-hematite exsolution pair. Secondary to this is very fine grained magnetite, with a few larger euhedral magnetite grains present, some contained within clusters of mica grains.

Discussion

Petrology. The Elberton pluton is a fine grained granite (classification of *Streckeisen* [1967]) composed of equal pro-

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TABLE 3. Remanent Magnetic Results for the Elberton Pluton

Sample	D	I	θ	ϕ	$J \times 10^{-2}$
EB-1-1*	307.5	38.1	42.4	185.9	0.05
EB-1-2*	205.0	−1.4	−49.3	236.6	0.02
EB-1-3	323.5	33.5	53.9	170.6	0.04
EB-1-4	352.5	14.7	62.6	113.3	0.11
EB-1-5	339.0	24.1	61.3	143.8	0.17
EB-1-6	7.5	15.9	63.2	80.3	0.23
EB-1-7	336.9	7.9	53.1	137.7	0.12
EB-1-8	351.4	33.6	72.6	125.3	0.18
EB-1-9	17.4	15.2	59.2	61.6	0.15
EB-2-1*	335.5	−9.6	44.8	132.6	3.04
EB-2-2	165.5	−21.9	−63.8	310.7	3.70
EB-2-3	156.6	−13.8	−55.4	321.0	5.69
EB-2-4	162.1	−19.0	−60.7	315.3	5.67
EB-2-5	160.8	−14.2	−57.9	314.8	5.91
EB-2-6	159.7	−15.3	−57.8	317.2	6.40
EB-2-7	158.8	−11.8	−55.8	316.8	8.03
EB-2-8	160.1	−7.4	−54.5	312.8	4.13
EB-2-9	167.1	−16.8	−61.9	305.0	8.15
EB-3-1	179.3	−12.5	−62.3	278.5	5.95
EB-3-2	178.8	−10.6	−61.3	279.5	5.10
EB-3-3	167.8	−8.0	−57.9	300.4	6.26
EB-3-4	173.8	−10.8	−60.9	289.8	4.48
EB-3-5	175.6	−9.5	−60.5	285.9	4.71
EB-3-6	168.2	−14.1	−61.0	301.8	7.91
EB-3-7	175.4	−11.4	−61.4	286.6	6.88
EB-4-1	354.2	9.4	60.2	108.7	6.09
EB-4-2	347.0	6.2	56.8	121.2	3.62
EB-4-3	343.9	6.5	55.8	126.5	3.84
EB-4-4	346.6	4.5	55.9	121.4	6.94
EB-4-5	349.6	4.4	56.8	116.2	4.76
EB-4-6	348.7	6.9	57.7	118.5	3.23
EB-4-7	346.2	10.0	58.3	123.9	4.76
EB-4-8	351.2	8.4	59.1	114.3	4.05
EB-4-9	347.2	10.0	58.7	122.1	6.05
EB-5-1*	61.4	24.6	30.8	12.1	0.53
EB-5-2*	176.3	−9.6	−60.6	284.5	0.74
EB-5-3	301.5	−1.2	25.3	167.6	0.59
EB-5-4	19.5	10.7	56.2	60.3	0.22
EB-5-5	323.0	23.1	49.8	162.8	0.12
EB-5-6	323.2	13.8	46.6	157.0	0.25
EB-5-7	9.9	−2.9	53.3	80.3	0.12
EB-5-8*	173.8	27.3	−41.2	285.0	0.25
EB-5-9	319.3	10.7	42.7	159.1	0.13
EB-6-1*	110.4	2.0	−16.2	354.4	0.29
EB-6-2	162.1	25.0	−39.9	300.0	0.23
EB-6-3	153.4	12.6	−42.4	314.1	0.24
EB-6-4	149.8	15.8	−39.1	316.9	0.18
EB-6-5	163.7	3.3	−51.2	303.6	1.01
EB-6-6	148.4	13.3	−39.5	319.4	0.29
EB-6-7	157.6	14.7	−43.4	308.3	0.36
EB-6-8	171.3	−14.5	−62.2	295.7	0.62
EB-7-1	167.4	−1.2	−54.6	299.1	1.53
EB-7-2	170.6	−7.0	−58.3	295.1	5.67
EB-7-3	166.4	1.2	−53.1	300.1	3.26
EB-7-4	165.5	−5.4	−55.9	303.5	1.79
EB-7-5	167.3	−2.9	−55.4	299.7	3.67
EB-7-6	172.7	−12.7	−61.6	292.4	2.96
EB-8-1	180.8	1.2	−55.4	275.6	0.84
EB-8-2*	334.0	−10.1	43.8	134.2	0.62
EB-8-3*	189.3	68.1	−4.4	271.2	0.56
EB-8-4	161.8	8.0	−48.3	304.9	1.18
EB-8-5	147.8	9.2	−40.9	321.6	0.24
EB-8-6	162.5	5.0	−49.9	304.8	1.04
EB-9-1	162.1	1.1	−51.6	306.6	12.50
EB-9-2	176.4	−0.4	−56.0	283.4	15.90
EB-9-3	162.3	−0.1	−52.2	306.7	7.85
EB-9-4	164.2	−1.6	−53.7	304.3	7.84
EB-9-5	163.3	3.3	−51.0	304.2	15.30
EB-9-6	162.8	−1.1	−52.9	306.3	10.20
EB-10-1	163.6	−8.0	−56.4	307.5	5.88
EB-10-2	163.9	−7.7	−56.3	306.9	3.15
EB-10-3	168.1	−8.9	−58.5	300.1	6.79
EB-10-4	162.2	−10.3	−56.8	310.8	5.96
EB-10-5	164.3	−0.4	−53.1	303.8	6.33
EB-10-6	169.1	−8.7	−58.7	298.3	5.96

TABLE 3. (Continued)

Sample	D	I	θ	ϕ	$J \times 10^{-2}$
EB-11-1	170.8	6.4	−51.8	291.9	3.25
EB-11-2	167.6	5.3	−51.5	297.2	5.73
EB-11-3	168.8	0.3	−54.3	296.4	2.26
EB-11-4	172.6	2.6	−54.0	289.6	3.27
EB-11-5	170.2	5.9	−51.9	293.0	2.97
EB-11-6	166.5	5.4	−51.2	298.8	3.00
EB-12-1*	42.2	40.8	51.8	11.8	2.39
EB-12-2*	114.6	−5.6	−21.8	355.0	1.02
EB-12-3*	191.5	18.2	−45.3	260.7	2.02
EB-12-4*	64.5	−13.0	16.9	27.4	0.04
EB-12-5*	169.2	−20.5	−64.6	302.4	0.11
EB-12-6*	201.1	23.3	−39.7	249.8	0.05
EB-13-1*	125.5	82.6	24.8	290.0	0.18
EB-13-2	344.8	−19.6	43.6	117.9	0.16
EB-13-3	333.0	−20.8	38.4	131.7	0.15
EB-13-4	318.9	6.6	41.0	157.4	0.09
EB-13-5	357.8	−10.3	50.8	100.4	0.18
EB-13-6	355.3	4.9	58.2	105.9	0.16
EB-14-1	169.3	−4.6	−56.8	296.8	4.99
EB-14-2	167.1	−2.2	−55.0	299.9	4.75
EB-14-3	165.0	−6.1	−56.0	304.5	6.75
EB-14-4	161.0	−3.9	−53.4	310.0	4.81
EB-14-5	167.6	−3.4	−55.7	299.4	6.52
EB-14-6	168.5	−2.0	−55.3	297.5	5.07
EB-15-1	344.2	−5.2	50.5	122.3	0.07
EB-15-2	355.5	5.9	51.5	138.6	0.14
EB-15-3*	190.2	1.2	−54.1	259.4	0.23
EB-15-4*	173.8	19.5	−45.6	285.7	0.20
EB-15-5	342.4	4.8	54.4	128.2	0.20
EB-15-6	338.2	−1.8	49.5	131.9	0.09
EB-16-1*	359.5	49.7	86.5	104.1	5.64
EB-16-2	340.7	4.3	53.4	130.6	6.74
EB-16-3	342.6	8.9	56.3	129.5	4.73
EB-16-4	319.4	5.4	40.9	156.3	6.69
EB-16-5	345.3	6.4	56.3	124.1	7.79
EB-16-6	342.3	9.4	56.4	130.2	6.30
EB-17-1	165.8	−4.0	−55.4	302.5	4.28
EB-17-2	168.0	−5.1	−56.6	299.2	4.62
EB-17-3	140.6	1.4	−39.3	332.1	7.18
EB-17-4	165.1	−5.0	−55.6	304.0	5.20
EB-17-5	165.3	−5.2	−55.7	303.7	5.39
EB-17-6	165.7	−11.6	−58.9	305.4	2.63
EB-18-1	166.2	−7.1	−57.0	302.9	2.87
EB-18-2	165.4	5.5	−50.8	300.4	2.76
EB-18-3	158.9	0.3	−50.5	311.5	2.46
EB-18-4	165.3	1.2	−52.8	301.8	2.27
EB-18-5	161.7	2.8	−50.6	306.6	1.69
EB-18-6	159.2	5.9	−48.2	309.1	2.40
EB-19-1	162.4	−3.9	−54.0	307.9	7.91
EB-19-2	166.7	1.8	−52.9	299.4	4.35
EB-19-3	164.4	−0.2	−53.1	303.6	5.20
EB-19-4	161.2	−3.9	−53.5	309.7	2.89
EB-19-5	163.5	−1.0	−53.1	305.2	4.98
EB-19-6	159.4	1.9	−50.1	310.2	6.46
EB-20-1*	345.9	−59.6	17.6	108.7	0.03
EB-20-2	15.2	−24.0	41.3	77.1	0.03
EB-20-3	341.5	−27.7	38.2	120.0	0.06
EB-20-4	12.8	−25.2	41.2	80.3	0.05
EB-20-5	349.7	−1.1	54.1	114.7	0.07
EB-20-6*	50.3	−58.1	3.6	60.0	0.10
EB-21-1	160.5	−5.3	−53.8	311.3	0.74
EB-21-2*	180.8	22.1	−44.5	275.9	1.62
EB-21-3	165.6	10.4	−48.5	298.9	0.83
EB-21-4	146.8	1.4	−43.4	325.8	3.24
EB-21-5	167.7	1.4	−53.4	297.9	1.75
EB-21-6	146.6	−6.0	−46.1	329.4	0.69

Sample is within site sample number; data reported are for samples demagnetized at 40 mT, which was chosen based on pilot sample demagnetization. D is RM declination; I is RM inclination; θ and ϕ are the latitude and longitude of the pole position, respectively, determined for each sample. J is intensity of magnetization after demagnetization in amperes per meter.

*Samples which were rejected from final analysis (Table 4) based upon anomalous directions.

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TABLE 4a. Elberton Pluton Site Means (40mT Results)

Site Number	N	$\overline{D}$	$\overline{I}$	θ	ϕ	Polarity	R	Kappa	α_{95}
EB-1-	7	350.2	21.6	65.5	120.6	N	6.6447	16.9	15.1
EB-2-	8	161.3	−15.0	−58.5	314.5	R	7.9673	213.8	3.8
EB-3-	7	174.1	−11.0	−61.0	289.1	R	6.9777	269.4	3.7
EB-4-	9	348.3	7.4	57.8	119.3	N	8.9830	470.5	2.4
EB-5-	6	335.6	10.2	53.3	140.6	N	5.2336	6.5	28.4
EB-6-	7	158.0	10.2	−45.7	309.2	R	6.7990	29.8	11.2
EB-7-	6	168.3	−4.7	−56.5	298.5	R	5.9761	208.8	4.6
EB-8-	4	163.3	6.0	−49.8	303.4	R	3.9120	34.1	16.0
EB-9-	6	165.2	0.2	−53.2	302.2	R	5.9742	194.0	4.8
EB-10-	6	165.2	−7.3	−56.7	304.6	R	5.9850	332.8	3.7
EB-11-	6	169.4	4.3	−52.5	294.5	R	5.9920	622.1	2.7
EB-13-	5	342.1	−8.1	48.3	124.5	N	4.7417	15.5	20.1
EB-14-	6	166.4	−3.7	−55.4	301.4	R	5.9912	570.8	2.8
EB-15-	4	340.1	0.9	51.6	130.3	N	3.9800	150.0	7.5
EB-16-	5	338.1	7.0	53.3	135.5	N	4.9304	57.5	10.2
EB-17-	6	161.8	−5.0	−54.2	309.2	R	5.9056	53.0	9.3
EB-18-	6	162.8	1.4	−51.7	305.5	R	5.9747	197.6	4.8
EB-19-	6	162.9	−0.9	−52.8	306.0	R	5.9899	497.0	3.0
EB-20-	4	359.7	−20.2	45.6	97.4	N	3.8213	16.8	23.1
EB-21-	5	157.4	0.4	−49.8	313.4	R	4.9110	45.0	11.5

Site is location from Figure 1; N is number of samples within each site; $\overline{D}$ is mean declination; $\overline{I}$ is mean inclination; ϕ, θ are pole longitude and latitude; α_{95}, R, and kappa, are Fisherian statistics [*Fisher*, 1953]; Tilt correction azimuth is the direction of structural rotation (see text), dip is the magnitude of that rotation (see text); the total number of sites is 20 (site EB 12 was rejected from consideration due to a within site $\alpha_{95} > 30°$); d_p is the error in determining the great circle distance of the pole from the sample site; *dm* is an error in declination which corresponds to a displacement from the pole, normal to the pole–sample site great circle. It is important to note that it has been suggested that samples with high percentage anisotropy (% An > 10%) may yield RM directions which are suspect (for example, *Stacey* [1960], *Girdler* [1961], *Uyeda et al.* [1963], and others). However, all but five sites from the Elberton pluton yield % anisotropy values which are less than 5% [*Ellwood and Whitney*, 1980].

portions of quartz, microcline, and oligoclase with several percent biotite and accessory allanite. Several different oxide minerals are present in minor proportions. The dominant species, an ilmenite-hematite exsolution pair, is apparently derived from original ilmenite-hematite and ferrianilmenite solid solutions. Much less common are nearly homogeneous magnetite grains with minor ilmenite intergrowths. Both of these phases are euhedral to subhedral, about 0.2 to 1 mm in size, and are probably derived from magmatic phases reequilibrated during cooling of the body. Also present is an extremely fine grained, opaque mineral within the alkali feldspar. This phase gives a dusty appearance to the feldspar and gives color to the granite. In all cases where this phase can be identified, it appears to be magnetite, although absolute identification is difficult due to the fine particle size. In places where these inclusions have been partly oxidized to hematite, the granite becomes pink in color. The transition between this pink phase and the grey phase is completely gradational across a few meters and probably reflects low temperature oxidation.

Crude estimates of ilmenite composition based on visual methods and a coexistence with ilmeno-hematite and magnetite suggest high oxygen fugacities, one to two log units below the hematite-magnetite buffer, and temperatures of crystallization around 650 to 700°C. Water contents of the magma are not known, but based on crystallization history and probable temperatures, values around 4% or so seem possible.

In summary, examination of polished and thin sections indicates that magnetite is present, is a premagmatic and very early postmagmatic mineral phase, and is probably the main RM carrier, recording both the geomagnetic field at the time of cooling and the crystal fabric developed during emplacement. The magnetic fabric is discussed elsewhere [*Ellwood and Whitney*, 1980].

Structural setting. The outline of the Elberton pluton as given on the geologic map of Georgia, was found to be in error. Areas shown as the Elberton pluton on the map contain, in addition to a fine-grained biotite granite, orthogneisses, paragneisses, amphibolites, metavolcanic rocks, and a major cataclastic zone (all unrelated to the Elberton pluton). Our mapping of the contact (Figure 1), which includes the work of *Rozen* [1978], and *Hess* [1979], decreases the size of the pluton by about 40%. Within the Elberton proper no internal contacts have been observed in any quarry or other site visited (with the exception of a biotite schlieren in one quarry). Petrographic and chemical analyses have also failed to show any major differences within this body. Therefore the Elberton as shown in Figure 1 appears to be a single intrusion and not a composite body.

The contact of the granite is often subhorizontal or shal-

TABLE 4b. No Tilt Correction

af Field	$\overline{D}$	$\overline{I}$	α_{95}	ϕ	θ	d_p	*dm*
40mT	344.4	2.2	3.1	124.1	54.0	2.2	4.5
80mT	348.1	1.1	5.1	118.0	54.7	3.0	6.1
Mixed	346.8	1.8	4.3	120.3	54.7	2.7	5.4

See notes to Table 4a.

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TABLE 4c. Tilt Corrected AZ = 102; Dip = 32

af Field	$\bar{D}$	$\bar{I}$	α_{95}	ϕ
40 mT	349	14	121	61
80 mT	352	12	114	61
Mixed	351	13	116	61

Amount necessary to bring magnetic foliation plane to horizontal.

lowly dipping with the Elberton lying below the country rocks. Thus the current erosion surface appears to be near the top of the body with many roof pendants present. The undulatory nature of the contact yields many apophyses and reentrants of granite making the outline of the body complex and often locally related to topography.

The Elberton has a faint foliation locally developed near its contacts. This fabric crosscuts all regionally developed tectonic fabrics in the surrounding rocks. The grain size decreases slightly near the contacts, a fact long recognized by local quarrymen, since the finer grained 'Blue Granite' is considered the best monument stone. Therefore the Elberton appears to be posttectonic and postmetamorphic with respect to the fabric elements developed in the surrounding rocks.

To the northwest the crustal block which includes the Elberton is bounded by a fault shown as the extension of the Towaliga on the geologic map of Georgia [*Lawton et al.*, 1976]. This fault has no apparent cataclastic fabric and appears to be delineated mainly on the basis of aeromagnetic data which show a linear series of small, magnetic anomalies oriented along it. In the field there are local lithologic discontinuities associated with this fault, but the cataclastic development often observed along the Towaliga bounding the Pine Mountain belt in southeastern Georgia is absent. We therefore refer to this fault as the Hartwell extension of the Towaliga, named after the town of Hartwell and the Hartwell dam near which the fault is located.

To the southeast the Inner Piedmont is terminated by the Middleton-Lowndesville cataclastic zone [*Griffin*, 1970, 1971, 1972; *Rozen and Whitney*, 1978; *Whitney et al.*, 1978, *Rozen*, 1978]. This zone has a well-defined cataclastic fabric. Fold structures within the cataclastic zone suggest a thrusting motion, accompanying the cataclastic period, of the Charlotte and Slate belt rocks over the Inner Piedmont units. The cataclastic rocks within this zone have subsequently been crosscut by aplite dikes which do not show any evidence of cataclastic fabrics but are themselves locally offset by brittle faulting. This second period of movement was normal with the Inner Piedmont being uplifted relative to the Charlotte and Slate belts.

To date, the Elberton has not been seen in contact with the Middleton-Lowndesville shear zone. No cataclastic development has been observed in the Elberton even where it is within one-half kilometer of the shear zone, although some slight plastic deformation features are observed. The aplites cutting the cataclastic zone are petrographically similar to those internal to the Elberton, and these in turn appear to be cogenetic with the granite. If these aplites are coeval with the granite, then Elberton emplacement postdates thrusting along the cataclastic zone but precedes normal faulting. Thus these observations imply a relationship (although somewhat tenuous) between the Elberton and the two stages of movement along the Middleton-Lowndesville zone.

Remanent magnetism. Since the Elberton granite is relatively fine grained and uniform throughout the area in which it outcrops, a large number of widely spaced, mineralogically similar and cogenetic samples are available for sampling. This has proven to be an ideal setting for studying the magnetization of this granite in terms of the genetic implications of the unit.

The af demagnetization. The af demagnetization of pilot samples (Table 2) shows that the paleomagnetic stability index (PSI) of *Symons and Stupavsky* [1974] is optimum over a range of 20–100 mT. This is indicative of a large range of magnetic coercivities and was expected from the iron oxide mineralogy. At the same time the RM directions measured, in general, vary only slightly with increased af demagnetization treatments. Based on these data, demagnetization steps of 40–80 mT were chosen for all samples.

The separation of magnetic components method of *Stupavsky and Symons* [1978] was employed in this study and indicates that for most Elberton sites the magnetization resides in at least three or more components (Table 2). This is in agreement with our in situ thermal demagnetization experiment (Figure 3; discussed later).

RM site means were calculated for each af demagnetization step (40 and 80 mT, respectively) and for a combination of the 40 and 80 mT data using the kappa maximization method of *Watkins et al.* [1972] (Table 4*b*). While it is reasonable to report only the data based solely on the kappa maximization approach, due to the large range of coercivities indicated, we have reported means for all the data for comparison. All means are very similar (difference 3.7° for mean declinations and 1.1° for mean inclinations) and are statistically indistinguishable at the 95% significance level (Table 4*b*).

RM polarities. Mean directions of RM and corresponding poles for all sites before a slight tilt correction has been applied are listed in Table 4*a*. It is apparent from an examination of these poles that several sites are normally magnetized and several are reversed (Figure 4). Some samples within sites exhibit opposite polarities (Table 3, Figure 5). Two explanations for two polarities within the same body exist: first,

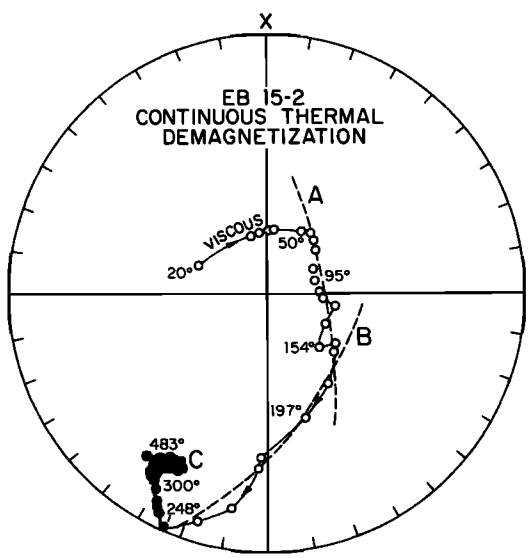


Fig. 3. Equal area projection of the change in magnetization for sample EB 15-2 during continuous thermal demagnetization (see text). Open symbols are plotted on the upper hemisphere; closed symbols on the lower hemisphere. Numbers are representative temperatures during demagnetization. See text for a discussion of great circle paths A and B, the small circle viscous change, and point C, the characteristic magnetization. Cross represents an arbitrary 0° declination (see text).

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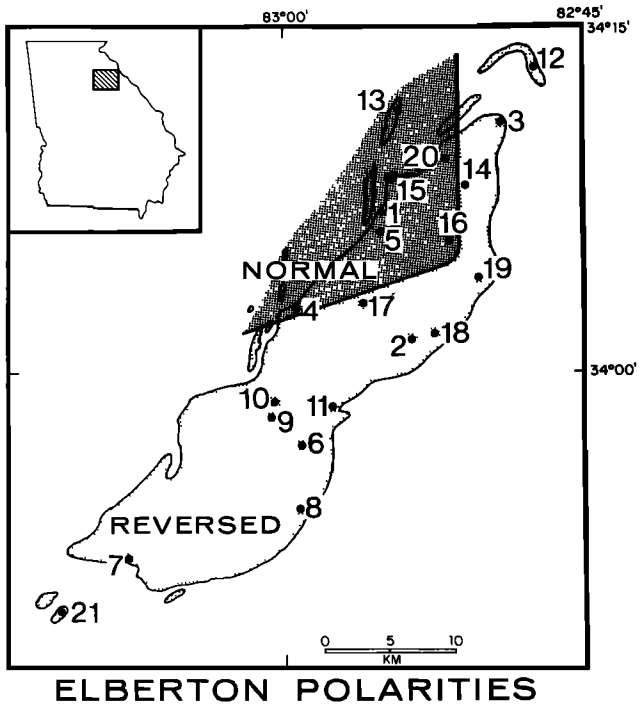


Fig. 4. Characteristic site polarities for the Elberton, taken from Table 4; normal polarity is dark stippled; reversed polarity is light stippled. Pluton boundaries are simplified from Figure 1.

the geomagnetic field reversed during cooling of the pluton with both polarities being recorded, or second, self-reversed samples (possibly whole sites) are present.

Self reversals. *Merrill* [1975] has summarized the early reported occurrences of self-reversals in igneous rocks. Much of this work was conducted by Japanese workers [for example, *Nagata et al.*, 1953] after the discovery of the self-reversed remanent magnetic property exhibited in the Haruna Dacite from Japan. Other analyses included observations of self-reversals in a granite-aplite dike [*Heller and Egloff*, 1974]. *Hoffman* [1975] has examined the mineralogical variations in ilmenite-hematite ratios shown to be responsible for the self-reversing phenomenon. It has also been shown that self-reversals can arise from low temperature oxidation [*Merrill and Gromme*, 1969].

Examination of the dominantly reversed versus dominantly normal sites in the Elberton yields the nonuniform polarity distributions illustrated in Figure 4. While there is a good between site polarity consistency (no site within a zone has a dominant anomalous polarity), there is no apparent relationship to cooling margins. Estimation of the cooling margins is based on the anisotropy of magnetic susceptibility data of *Ellwood and Whitney* [1980], which indicate a general symmetry within the zone of exposure of the pluton. If the polarities exhibited were acquired as a result of a change in polarity of the geomagnetic field, then a polarity symmetry which mirrors the cooling interfaces would be expected as a result of a migrating Curie point isotherm. Even within single sites which have recorded both polarities there is no symmetry relative to polarity zone margins (Figure 5). Such discrete, nonuniform pods may indicate that one of the polarities in the Elberton pluton is the result of a self-reversing mechanism. This may be due to the presence of very low percentages of fine-grained magnetite (usually the major RM carrier) being dominated by a stable RM carrier of ilmeno-hematite in the self-reversed composition range (despite the dominance of magnetite at high fields in the thermomagnetic balance) or to some other mineralogical phenomenon such as low temperature oxidation, discussed by *Merrill and Gromme* [1969]. As a check on the behavior of the RM intensity between different polarity samples at site 15, we thermally demagnetized, in air, two normal (EB 15-09, −10; Table 5) and two reversed (EB 15-11, −12; Table 5) samples. Results show that for normally magnetized samples, while the RM direction remained normal and appeared to be stable through ~700°C treatments, the normalized intensity fell to values of ~0.1, and the median destructive temperatures exceeded 400°–500°C (Table 5). Reversed samples, on the other hand, started with initial directions representing present field directions and reached a reversed direction after thermal demagnetization above 500°C. Such a normally magnetized overprint would be expected for samples with a magnetite carrier. Median destructive temperatures are 400°–500°C for these samples as well. The presence of a stable and consistent magnetization, after thermal demagnetization to 600°C or 700°C, indicates the presence of a high Curie temperature component of magnetization which has recorded a magnetization in the same direction as that carried by lower temperature components. We assume that hematite is the high temperature magnetic carrier and that since the hematite observed is probably magmatic or near magmatic in origin, the high temperature magnetization represents an RM acquired at the time of emplacement of the pluton.

A further demagnetization test was performed on normally magnetized sample EB 15-2. The sample was continuously

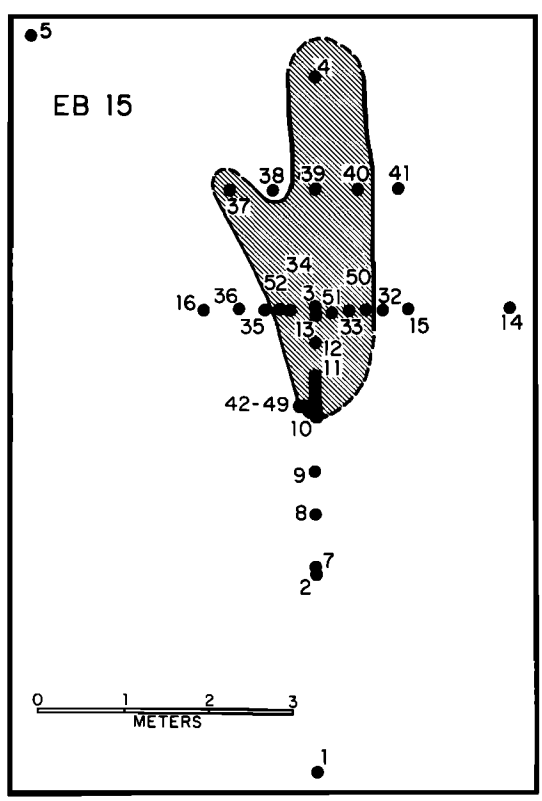


Fig. 5. Shaded area represents a reversed polarity zone at normally magnetized site EB 15. Sample numbers are given. Some samples (normal) lie outside the map area. Extra field sampling was required to define this pattern.

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TABLE 5. EB 15 Thermal Demagnetization Results

	H	D	I	J/Jo
	Normally Magnetized Samples			
EB 15-09	0	348.6	43.9	1.000
	100°	353.2	45.1	0.912
	200°	346.1	42.7	0.745
	300°	349.2	44.4	0.638
	400°	338.5	41.7	0.478
	500°	339.0	40.3	0.366
	600°	352.6	48.6	0.212
	700°	38.0	48.2	0.082
EB 15-10	0	0.0	46.8	1.000
	100°	358.2	50.7	0.929
	200°	355.4	47.5	0.812
	300°	355.0	48.3	0.700
	400°	356.5	48.5	0.558
	500°	338.7	31.9	0.485
	600°	340.1	27.0	0.295
	700°	342.9	52.5	0.077
	Reversed Samples			
EB 15-11	0	21.0	73.0	1.000
	100°	35.9	78.3	0.946
	200°	62.4	79.2	0.818
	300°	90.1	77.2	0.725
	400°	128.0	70.7	0.606
	500°	158.5	62.7	0.533
	600°	156.9	33.8	0.426
	700°	114.3	38.5	0.125
EB 15-12	0	0.5	76.0	1.000
	100°	354.9	72.0	0.973
	200°	11.8	79.1	0.852
	300°	29.9	81.0	0.758
	400°	100.4	77.2	0.648
	500°	134.0	74.8	0.424
	600°	164.1	40.1	0.413
	700°	152.7	46.2	0.187

H is demagnetization temperature in °C; *D* is declination; *I* is inclination; *J/Jo* is normalized intensity.

measured in a cryogenic magnetometer during thermal demagnetization. Relative changes in magnetization were monitored and are plotted in Figure 3. While it was not possible to orient the sample with respect to the horizontal plane, a relative coordinate system is used in Figure 3 (cross corresponding to an arbitrary 0° declination) to illustrate the RM changes within the sample. Since the vertical orientation of the sample has been preserved, we infer the removal of three components of magnetization; the first, a viscous normal magnetization (small circle movement), was removed by temperatures of 50°C; the second (A in Figure 3), a normal magnetization, was removed after demagnetization to ~160°C and the third (great circle path B in Figure 3), another normal magnetization, was removed after demagnetization to ~300°C. The remaining magnetization (C in Figure 3), a characteristic reversed magnetization, did not change direction with demagnetization to 483°C at which point vacuum loss due to sample degassing prevented demagnetization at higher temperatures.

In summary, thermal demagnetization of normally magnetized samples showed very little or no acquired viscous magnetization relative to reversed samples, and no reversed component was evident when these samples were allowed to cool before RM remeasurement. If, on the other hand, samples are not allowed to cool before RM measurement (as in the case for EB 15-2), then removal of a normally magnetized overprint is evident and a reversed component becomes dominant at high temperatures. We interpret these results to indicate that (1) a self-reversed weak RM is carried in the ilmeno-hematite, but (2) becomes apparent only when the magnetite content is so low that the RM carried in the ilmeno-hematite dominates. This suggests that the geomagnetic field was reversed when the Elberton granite acquired its magnetization and the normally magnetized samples result from a self-reversing mechanism.

Geomagnetic field reversals. It has been shown that deep seated intrusives can record reversals of the earth's geomagnetic field [*Dodson et al.*, 1978]. If the RM reversals in the Elberton represent reversals of the earth's magnetic field and not self-reversals, it should be possible to evaluate the cooling history of the granite using the RM transition zone between normally magnetized and reversely magnetized samples within the same site.

Metamorphic slow cooling model: If the Elberton Granite was emplaced within high temperature country rock and cooled only very slowly as the entire area cooled, then we can test this model based on the following assumptions:

1. The granite cooled below the blocking temperature of magnetite (~580°C) at ~350 m.y. B.P., this is based on the Rb/Sr age, the RM data generally consistent with a 350 m.y. age for the granite, and on Curie temperatures for Elberton samples.

2. The granite cooled below ~350°C by ~250 m.y. B.P., based on estimates for argon and strontium retention ages in mica discussed earlier.

3. From assumptions 1 and 2 we conclude that a cooling of ~230°C occurred over a period of 100 m.y.

4. If we assume that the geothermal gradient was 25°C/km (based on heat flow data for a similar tectonic province [*Lachenbruch and Sass*, 1977]) this will yield an uplift for the Elberton of ~9 km in 100 m.y. This, in turn, yields a cooling surface migration rate of 0.09 mm/yr.

5. If it is further assumed that the geomagnetic field fully reverses over a period 1000 to 5000 years, then such a reversal will be recorded in the Elberton over a minimum ground distance of 9 to 45 cm.

At site EB 15 the reversal pattern is very well defined, lying between samples which are only a few centimeters apart (Figure 5). Therefore while a reversal recorded over a distance of 9 to 45 cm does not necessarily exclude such a slow cooling model as the basis for the reversal pattern, the data are not in full agreement with such a model.

If the body remained at high temperatures for a very long period of time, one might expect alteration and secondary mineral growth in the granite. Since the granite is relatively fresh, this may indicate that either alteration did not occur because of a lack of water in the magma or that the slow cooling model is not viable.

Magmatic quick cooling model: Another model which can be tested is based on the assumption that the Elberton pluton was emplaced within an already reasonably cool country rock, and cooling was a function of local heat loss.

If we combine *Jaeger*'s [1969] equations 9 and 10, we arrive at

$$T = kt/a^2 \quad (1)$$

where a is the half width of the body (5×10^5 cm), k is the thermal diffusivity (0.01 cm^2/s), and t is the time since intrusion. By solving (1) for the time at which the 600°C isotherm (the approximate temperature for RM acquisition) reaches the

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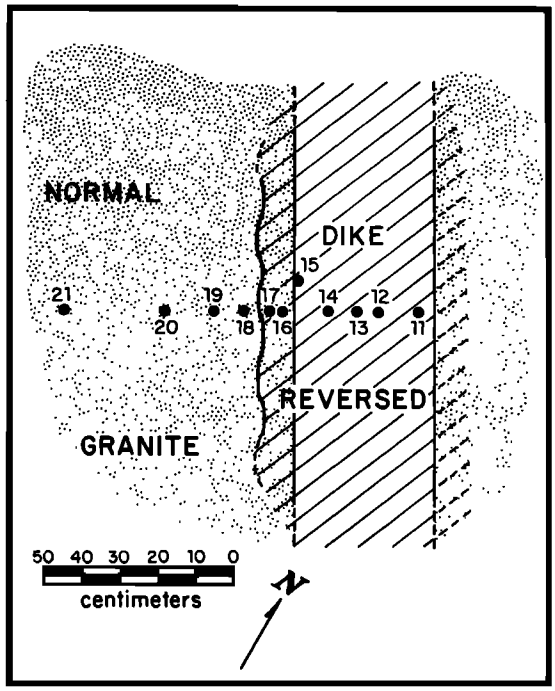


Fig. 6. Reversed basalt dike (shaded) in a normally magnetized quarry (stippled; site 4). Cross hatch lines indicate zone of reversed magnetation, the zone including granite samples 16 and 17. Sample locations and sample numbers are shown and correspond to numbers reported in Table 6.

center of the body (assuming $T = 1$ for an initial magma temperature of 800°C and an initial country rock temperature of 400°C), we obtain a time of 25×10^{12} s (~800,000 years). Assuming 5000 to the center of the Elberton, the rate of movement of the isotherm is 6.25×10^{-3} m/yr. Again if a reversal requires between 1000 and 5000 years, it would be recorded over a ground distance of 6 to 30 m. This is not the case for site 15 (Figure 5), which exhibits a reversal over a much shorter ground distance.

The basic assumptions involved in estimating the distance over which a reversal will be recorded within this pluton can be varied over a large range. The basic difference, however, between a self-reversal, the metamorphic slow-cooling model, or the igneous quick cooling model, that is, very short versus short versus long RM reversal distance, has been resolved with the RM data, which we interpret to indicate a self reversal in the granite. The metamorphic slow-cooling and the magmatic quick cooling models are excluded.

Baked contact stability test. Examination of the quarry at site 4 revealed the presence of a diabasic intrusion into the granite and provides an opportunity for a baked stability test. In principle, an intrusion into the pluton should have reheated the granite immediately adjacent to the dike/granite contact, and upon cooling the reheated granite should have acquired the geomagnetic field direction at the time the dike was intruded. Thus if the normal polarity component does not reside in the baked zone of the granite, it would indicate that this component was acquired before intrusion of the dike. Figure 6 illustrates RM results for cores taken at close intervals within the dike and in the granite immediately adjacent to the dike. Numerical results after af demagnetization are given in Table 6. The dike has acquired a reversed and stable RM direction. It is apparent that the granite samples in close proximity to the dike have acquired nearly the same reversed direction while the rest of the granite samples, farther from the contact, yield the normal RM direction consistent with other nonreheated samples from site 4 (Table 4). We interpret these data to indicate that the reheated granite has acquired a remanence which appears to be stable over geologically long time periods. If the unheated granite polarity at site 4 results from a self-reversed magnetization, then the reversed polarity reheated granite samples 4-16 and 4-17 (Figure 6; Table 6) would appear to be anomalous. We infer from this that either (1) the reheated granite no longer exhibits a self-reversed magnetization, possibly due to a change in magnetic mineralogy during reheating or (2) that the high intensity of magnetization of the high blocking temperature diabase, in close juxtaposition to granite samples 16 and 17, may dominate over the lower intensity, high blocking temperature hematite magnetization assumed to be responsible for the self-reversal.

It is interesting to speculate that while the dike has not been independently dated, it appears very fresh in thin section and yields a single site pole coincident with *Irving*'s [1979] apparent polar wander path at ~200 m.y. We suggest that the dike may have been intruded into the Elberton granite in late Triassic time and is one of many such dikes found intruded into rocks along the eastern margin of North America.

Paleomagnetic age estimates. Figure 7 is a northern hemisphere equal area projection on which is plotted the apparent polar wander path for North America of *Irving* [1979] for the period 350 m.y. to 150 m.y. B.P. Also plotted are two pole positions, with confidence limits, for the Elberton pluton. Point A represents the Elberton pole before a correction for physical rotation of the body (Table 4*b*). B is that pole after a structural correction based on the anisotropy of magnetic susceptibility (AMS) results reported by [*Ellwood and Whitney*, 1980] and the assumption that the magnetic foliation plane was initially horizontal and has since been rotated into the present orientation (Table 4*b*). Since confidence limits for A and B overlap with each other and with a 10° cone of confidence for the 350 m.y. paleopath of *Irving* [1979], these data are consistent with the 350 m.y. Rb/Sr whole rock age. A and B also overlap with the confidence limits for a 245 m.y. paleopole. A 245 m.y. age therefore is not excluded by interpretation of the paleomagnetic data.

Point B (rotation based on the assumption of an initially horizontal magnetic foliation plane) is located farther from the polar path than is point A (no structural correction). Since *Irving*'s [1979] 350 m.y. paleopole position is poorly defined, the increased separation of B from the polar path may only be

TABLE 6. Site 4 Baked Contact Stability Test

Dike			Granite		
Sample Number	*D*	*I*	Sample Number	*D*	*I*
11	181.1	−18.5	16A	196.2	−14.6
12	187.6	−17.9	16B	193.8	−7.8
13	184.1	−14.0	17	206.6	−16.0
14	183.8	−16.4	18	348.2	7.2
15A	189.3	−19.5	19	348.0	3.7
15B	186.9	−23.0	20	348.3	3.1
			21A	346.5	5.1
			21B	347.1	6.2

Sample numbers illustrated in Figure 6, for dike and granite samples drilled at site 4. *D*, *I* = RM declination and inclination after af demagnetization to 30 mT. A and B are specimens from the same sample.

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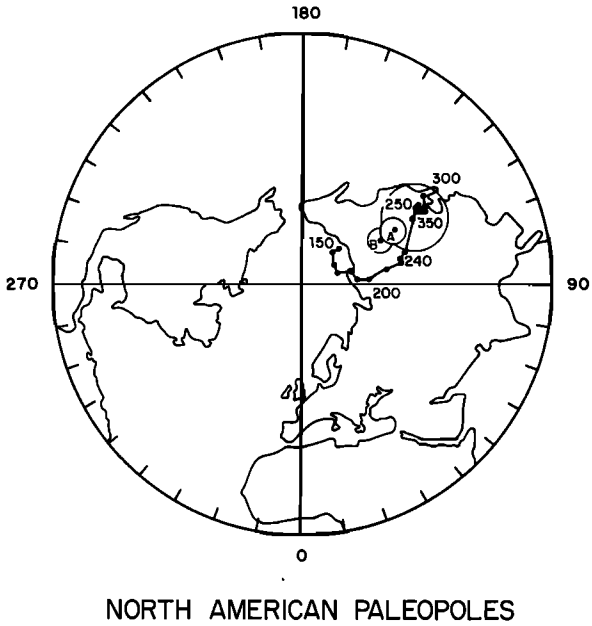


Fig. 7. North polar equal area projection including the North American polar Wander Path of *Irving* [1979] from 350 to 150 m.y. B.P., A, and B = Elberton poles, nontilt corrected and tilt corrected, respectively, using the AMS magnetic foliation plane method (see text).

fortuitous. However, if one assumes that the 350 m.y. paleopole reported by *Irving* [1979] is correct, then a physical rotation of the Elberton, to bring A into coincidence with Irving's paleopole, of ~15° down to the NW about a E-NE strike is required. While the axis of rotation is not too different from that defined by the AMS method, the tilt direction is in the opposite sense. Both tilt magnitudes are relatively slight.

Paleomagnetic results for Upper Devonian and Lower Carboniferous rocks from North America have been reported and summarized, for example, by *Kent and Opdyke* [1978], *Van der Voo et al.* [1979], *Kent* [1979], and *Scott* [1979]. These results are similar to the 350 m.y. paleopole reported by *Irving* [1979], although pole positions from Arizona given by *Elston and Bressler* [1977] have steeper pole latitudes similar to our B (after AMS tilt correction), and our A falls very close to the Upper Devonian apparent polar wander path of *Van der Voo* [1979].

We suggest two interpretations for the data. First, the structural reorientation of the body based on the assumption that the magnetic foliation plane was originally horizontal may not be valid and may indicate that the magnetic foliation plane was not originally horizontal. Second, a tilt correction (if any) will be of low magnitude, indicating that no major relocation of the granite is necessary to produce a pole position consistent with the 350 m.y. Rb/Sr whole rock age.

Elberton genetic model. The 350 m.y. old Elberton granite, situated along the COCORP seismic traverse, can ideally provide an indication of the age of emplacement of the Southern Appalachian allochthon. If the pluton is autochthonous, then the age of the intrusion places a minimum limit on the thrusting event of 350 m.y. B.P. An autochthonous Elberton pluton is consistent with the apparent in situ acquisition of a characteristic remanent magnetization, although a thrust model for the Elberton is not excluded by the paleomagnetic data. An autochthonous model, however, is also consistent with the structural relationship between the Elberton and the Middleton-Lowndesville thrusting previously discussed.

Further, *Wenner and Whitney* [1979] have shown, using the $^{18}O/^{16}O$ ratios in late orogenic granites from Georgia and South Carolina, that (1) ^{18}O-enriched granites (such as the Stone Mountain granite in Georgia) are confined to areas of regional gravity lows indicative of a deep continental crustal structure and are characteristic of a continental crustal protolith, (2) relatively ^{18}O-depleted granites to the southeast (such as the Siloam, Danburg, and Sparta granite plutons in Georgia), which occur in regions characterized by regional gravity highs and presumably indicative of a deep mafic crustal structure, suggest a mafic protolith, and (3) anomalous ^{18}O-enriched granites (such as the Clouds Creek granite in South Carolina) originated from what have been interpreted [*Long*, 1979] as isolated continental crustal fragments that are inferred from small-scale gravity lows. We interpret the gravity data to indicate that the Elberton granite lies adjacent to a major subcrustal boundary separating less dense continental rocks from more mafic crustal materials. This interpretation is consistent with the intermediate range of $\delta^{18}O$ values exhibited by the pluton.

It is apparent that an excellent correlation exists between the patterns exhibited by the $^{18}O/^{16}O$ data for these late orogenic plutons and the regional gravity data of *Long* [1979] which define deep, subcrustal lithologies. This consistency suggests that any large-scale thrusting involving the Piedmont probably predates intrusion of the Elberton granite. Any tilting that may have occurred has entailed only a very slight rotation about a horizontal generally NE-SW axis and is compatible with the Elberton pluton being contained within a hinge block separating a rising Inner Piedmont [*Dallmeyer*, 1978] from the relatively stable Charlotte-Slate belts. In Figure 8 we present a model which illustrates how the uplift of the Inner Piedmont may be absorbed by slight rotation of the Elberton hinge block and by shearing along the preexisting Middleton-Lowndesville zone and the concurrently developing Hartwell extension of the Towaliga fault zone. Sense of rotation may have been either to the southeast or northwest, but the magnitude was very small.

Significance for southern Appalachian tectonics. We infer from our results for the Elberton granite that no major post 350 m.y. compressional event has affected the pluton, either through physical translation or chemical alteration. This suggests that the Southern Appalachian region of the North

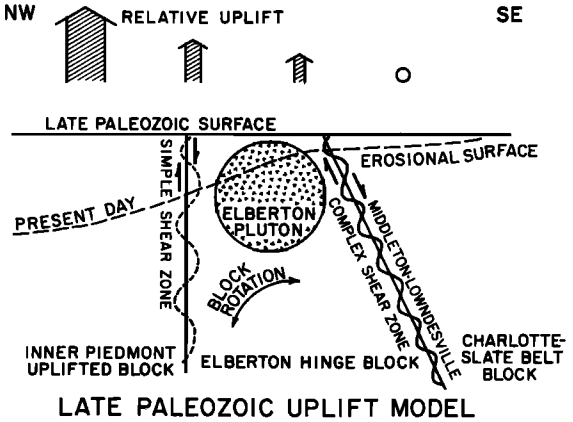


Fig. 8. Late Paleozoic uplift model of the Elberton hinge block showing possible slight postemplacement tectonic rotation.

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American continent has not been involved in a major continental collision in the last 350 m.y. Hercynian deformation (~270 m.y.) is present, however, in rocks farther to the southeast [*Snoke et al.*, 1980]. The lack of 350 m.y. and younger deformation in the Inner Piedmont and presence of 270 m.y. deformation to the southeast is more consistent with strike slip motion of the North American plate past some other continental boundary during the Carboniferous and Permian periods than with direct collision during the Permian. Continental reconstructions which are consistent with such motions have been presented by *Irving* [1979] and *A. G. Smith et al.* [1979]. We infer from our data that if the Inner Piedmont is part of a major allochthonous sheet (as suggested by *Cook et al.* [1979]), major episodes of movement along this sheet must have ended in Late Devonian time.

SUMMARY

Based on chemical, petrological, and remanent magnetic analyses for samples from 21 sites within the Elberton pluton, located in northeast Georgia, we conclude the following:

1. The Elberton granite yields a Rb/Sr whole rock age of 350 ± 11 m.y.

2. The body acquired a stable magnetization in the presence of an apparently reversed geomagnetic field. This field direction has been accurately recorded within the granite and is recoverable.

3. Some parts of the pluton are normally magnetized. Our preferred interpretation is that this is due to the granite exhibiting a self-reversed magnetization, which results from relatively low magnetite content and a self-reversed magnetization carried in the dominant (by volume) ilmeno-hematite magnetic phase present in the rock. We cannot exclude the possibility that the granite cooled very slowly as a result of intrusion into high temperature country rock and has recorded a reversal of the geomagnetic field. However, the presence of isolated small nonsymmetrical pods of one polarity within another suggests that this interpretation is not correct.

4. The Elberton has a Late Devonian-Early Mississippian paleomagnetic age (~350 m.y. B.P.), although a Late Permian magnetic age (approximately 245 m.y. B. P.) also fits the data.

5. While two possible tilt corrections might be inferred from the data, both are of slight magnitude and are not required to render the Elberton pole consistent with established apparent polar wander paths. Therefore we conclude that a major physical relocation of the granite is not necessary to produce a paleomagnetic age consistent with the available Rb/Sr whole rock age of 350 m.y. B.P.

6. We conclude from these data that no major continental collision involving the Southern Appalachian region of North America has occurred in the last 350 m.y.

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